

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY**MARCH/APRIL****ELECTRIC TRACTION****(MODEL QUESTION PAPER)****(ELECTRICAL & ELECTRONICS ENGG.)****(Maximum marks: 100)****[Time: 3 hours]****PART – A****Marks**

- I. Answer the following questions in *one or two* sentences. Each question carries 2 marks.
1. State any four elements of power supply system in electric traction.
 2. Write the term Encumbrances.
 3. Write any two drawback of pole collector.
 4. State the main methods for met the requirement of power in train lighting.
 5. Write any two advantages of search light signal. (5x2=10)

PART-B

- II. Answer any *five* of the following. Each question carries 6marks.
1. Explain sub sectioning and paralleling post in power arrangement.
 2. With neat sketch explain the 25KV overhead equipments (OHE) is supported by the bracket assembly swiveling type.
 3. Describe the factors determine the span length of overhead equipments(OHE).
 4. Explain the purpose of using pan head springs of high stiffness.
 5. Describe the advantages and disadvantages of over head system over third rail in electrical traction.
 6. Explain different types of signals and their positions.
 7. Explain various methods to achieve constant output in train. (5x6=30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT-I

- III. (a) Explain the working of remote control centre in electric traction system. (7)
 (b) Discuss merit and demerit of single unit versus multi unit substations. (8)

OR

- IV. (a) Explain the rating and tests capable of withstanding a circuit breaker used in electrical traction substation. (7)
 (b) Explain with the aid of sketch of substation equipment layout, the precautions taken to ensure the continuity of supply to OHE during fault or normal maintenance of substation.(8)

UNIT-II

- V a) Describe the difference between insulated overlap and uninsulated overlap in OHE. (7)

(b) Explain the neutral section in OHE. (8)

OR

VI (a) With neat diagram explain the construction of stitched catenary. (7)

(b) Write the use of aluminum and copper for OHE. (8)

UNIT-III

VII (a) Explain the advantages of single ended faiveley type pantograph over conventional diamond type pantograph. (7)

(b) Explain the working of pole collector and its advantages and disadvantages. (8)

OR

VIII (a) Explain the working of articulated system in pantograph. (7)

(b) Explain the phenomenon of critical speed. (8)

UNIT-IV

IX (a) Explain the centralized traffic control (CTC) and its advantages. (7)

(b) Explain the construction of search light and its advantages & disadvantages as compared to long range signal. (8)

OR

X (a) Explain with the aid of a diagram the working of double battery parallel block system. (7)

(b) Explain the mimic diagram and its supervisory facilities for controlled station. (8)
